Antagonist Zoologist Group

Effort by Gull

ZOO506 IMPORTANT MCQ

1. Bees. Wasps. Ants are included in order of class hexapoda ._____Hymenoptera

2. The length of period for effective treatment is more dependent upon the_____all

3. Insects belong in the Phylum_____Arthropoda

4. Beetles and Weevils are included in order. of class Hexapoda _____coleoptera

5. Centipedes are the member of class___Chilopod

6. Dragonfly are included in order of class hexapoda _____odonatan

7. Whiteflies have mouthparts._____sucking

8. Mosquito are included in order. of class hexapoda_____diptera9. Hyslop estimated in 1938

GULL

that the annual loss to agricultural crops by insects amounts to

somewhat over million dollars

_____

__1.600

10. Larvae of flies are usually called

_____Maggot

s

11. After the completion of the construction of adult structures the skin splits and sets free the

insect.

_____

_____Pupa

12. ….Transmission is mostly caused insects that have sucking mouthparts._____Biological

13. African sleeping sickness is caused by the vector.

_____Tsetse fly

14. Which of the following pathogen is carried by the whitefly to cotton? _Gemnivirus particle

15. In an average year loss to all sorts of crops by insect damage is about per cent_____10

16. A larva transforming in tunnels in wood forms a _____partial cocoon

17. Which of the following is NOT the stage of holometabolous life cycle?_____Nymph

18. In the… the insect hatches from the egg in a form totally unlike the adult____Holometabola

19. The disease rice dwarf virus is caused by the pathogen_____DSRNA particle

20. Insect vector for potatoplants_____Beetle

21. Insect vector for tomato plant is_____Whitefly

22. Which of the following insect vector causes Encephalitis?_____Culex mosquito

23. In the, . threads attach the hinder end of the body form 4 supporting loop around its middle._____Butterflies

24. In some insects maturity of the organs is not complete until a little later._____Reproductive

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25. Mango sudden death disease is caused by_Bark beetle

26. Typhus is caused by the insect vector_____Human louse flies

27. Phylodi disease in sesame indicum is caused by the pest vector_____Leaf hopper

28. Insect vector for banana trees is _____ aphid

29. Amoebic dysentery is caused by the vector, _____House fly

30. In case of transmission the pathogen is simply carried extremally or internally by the insects_____Physical

31. Sucking insects are the agents of transmission. _____biological

32.Mouth of neuropteran_____ _____Downward

33. Coal tar is a

_____

_____Repellent

34. How many types of Artificial control of insects?

_____6

35. Often some of the bug; body segments are fused with the head to form a crustacean_____

_____

_____Cephalothorax

36. The wings are_____outgrowths from the body._____Chitinous

37. _____temperature is not effective for killing of insects._____Low

38._____attacks canes.

_____Grassho pper

39. _____is a insects borne disease.

_____All

40. Bollworm makes_____on boll.

_____Bore hole/hole

GULL

41. Growing conditions for the insects is optimize_____control.
_____Cultural

42. Introduction of a foreign pest is a
_____Cul
tural

43. The_____extend from to body.
_____Elementary canal

44. Those Organisms which lay egg hatching later is called_____Oviparous

45. In the_____the young insect on escaping from the egg, though resembling its parent.

_____
____Hemimetabola or Paurometabola

46. Electrical window screens are available for
_____insects._____Electrocuting

47. Adult rice ear are_____in color.
_____Yellow

48. Natural Breeding method is a_____type of control._____Biological/Cultural

49. Bug worm is an example of _____Lepidoptera

50. Shellac is a_____product. _____Commercial

51. The_____are land animals breathing by air tubes opening on the sides of the body. _____Diplopoda

52. In the_____the segments of the body are grouped in three distinct sections.____Hexapoda

53. In_____mouth parts, being the more primitive and simple._____Biting

54. The ingrowth from the mouth is usually called the_ _____Fore intestine

55. An enlarged portion of the vas deferens is often present, for the temporary storage of the sperms, and is termed the_____ _____Seminal vesicle

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56-The… …extends from the mouth to the anus at the hinder end._____Alimentary canal

57-In insects Antenna are present in .....region.
_____Head

58-In some insects,egg remain in the body of the parent for a time after hatching are

termed.._____
_____ovoviviparous

59-Which type of skeleton is found in arthropod?. . _____Exoskeleton

60-In nervous system the cell bodies of neurons gathered together in clusters which are

called. _____
_____Gangila

61-The first abdominal segment is called the median segment or…_____Ropodeum

62-Near the base of the labium on its inner side there is a tongue like structure covered by bristles or

hairs
called…_____
_____Hypopharynx

63-Insects have ….
symmetry._____
_____Bilateral

64-Insects have …..pair/pairs of antennae on their head.
_____One

65-The ....is usually small and may not be visibleon all sides of the leg._____Trochanter

66-Which of the following is NOT the function of cerci?..._____used to store eggs

67-The sclerites on both sides of the segments present mainly in thoracic region are
called..._____
_____Pleuron

68-The stomach may develop side pouches called..
_____Gastric caeca

69-The nervous system of insects is located along the…..line of the body_____Middle

70-The in growth of the anus is called..
_____Hind intestine

71-In … …insect hatches from the egg practically in an adult condition._____Ametabola

72-Shell gland also opens into..
_____Seminal receptacle

73-When the body segments are fused with the head they form.._____Cephalothorax

74-The walls of Malpighian tubes consist of a single layer of cells surrounding a central channel

which at one end open into
the…_____Hind intestine

75-In the females insect the egg produced in a pair of ....located in the upper part of the

abdomen._____
_____ovaries

76-The ....are slight groove or scratches on the surface forming the movable joints._____sutures

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77-The ....is a tube-like organ at the end of the abdomen and is used for the laying of eggs.

_____
_____ovipositor

78-In adult insects, number of segments in thorax region is…_____3

79-Shedding of cuticula to let the organism grow is called..._____Molting

80-..are very rarely present on the abdomen in adult insects but are often found in the earlier

stages_____
_____legs

81-system of blood vessels coming to an end in ……region._____Abdomen

82-members of the ….. undergo gradual and usually incomplete metamorphosis. ___

_____
_____Paurometebola

83-The group of insects that supply nourishment to the… … before its birth are termed viviparous.

_____

_____young

84-In insects certain amount of poison can be eliminated from the body by

the…_____

_____Malpighian tubes

85-which of the following insect has opithognathous head?.._____Aphids

86-The .... forms the stomach of the adult insect._____mid intestine

87-In female insects, ovaries are located in the upper front part of the .._____Abdomen

88-The sclerites on both sides of the segments present mainly in thoracic region are

called..._____

_____pleuron

89-In some flies the larva shrinks within its larval skin and transforms,this skin is

called..._____

_____Puparium

90-total number of segments in the insects embryo is…._____21

91-which of the following is the part of hind intestine?..
_____Rectum

92-In ….heads ,labrium is present above the mouth opening._____prognathous

93-.. …is the respiratory structure in insects.
_____Book lungs

94-The compound eyes are also called …..eyes.
_____ocelli

95-Which of the following part of the mouth is concerned with the sense of
smell?.._____
_____Maillary palpus

96-Salt and ….are transported into the lumen of malpighian tubule from hemolymph._____uric acid

97-A pair of jaws present behind the labrium is called…
_____Madible

98-The hind intestine is derived into different parts which are… … colon and rectum_____ileum

GULL

99-..generally lie in the front of the thorax and open into the mouth_____Salivary gland

100- The tip of the wing is called

..____R

adiate

101- And the skeleton in arthropods is made up of …. Which somewhat resembles horn in its nature

_____

_____chitin

102. which of the following insect has opithognathous head?.._____Aphids

103. In insects Antenna are present in….. region.

_____Head

104. Tiny antenna- like structure consisting of one to six segments and attached to the maxilla is called

._____

_____Maxillary palpus

105. The …. are slight grooves or scratches on the surface forming the movable joints_____sutures

106. The number of ocelli in insects is

_____

__1-3

107. The size of ocelli is equal to the one facet of eye.
_____Compound eye

108. The front lip is called
_____
_____Lebrum

109. The hinder lip is called
_____
_____ Lebium

110. There are segments in the abdomen of the adult insect. _____ 3-11

ZOO506

IMPORTANT SHORT QUESTION

Q. Damage cause by termites.

They enter through cut ends of setts & feed on the soft tissue. The tunnel is filled with the soil.

The termites attack setts, shoots, canes and also stubbles Entire shoot dries up and can be pulled out easily

Q. Intelleigent cultivation.

- Intelligent cultivation and use of fertilizers is well as pruning to remove injured or diseased parts will aid in insect control

Q.what are repellents? Explain some repellents.

1-Repellents are those materials used to keep insects away from crops, animals and man

2. Various coal-tar and pine products are rather general insect repellents.

3. Naphthalene is a clothes moth repellent and oil of citronella is a common ingredient of preparations used to keep mosquitoes away from people

Q .write commercial products produce by insects..

A third group of insects is responsible for a number of commercial products

1-Silk

2-Honey

3-Beeswax

4-Shellac

Q. Write the uses of pesticides.

Pesticides are used to control various pests and disease carriers, such as mosquitoes, ticks, rats and mice. Pesticides are used in agriculture to control weeds, insect infestation and diseases. There are many different types of pesticides; each is meant to be effective against specific pests

GULL

Q. Mode of damage of fruit borer?

Bolls showing regular, circular bore holes Larvae seen feeding on the boll by thrusting their heads alone inside and leaving the rest of the body outside

Presence of granular faecal pellets outside the bore hole.

A single larva can damage 30-40 bolls

Q. Write the name of thorax seqments.

The thorax has its three segments usually quite clearly marked 1-Prothorax

2-Mesothorax

3-Metathorax

Q. Vector of malaria?

In the case of malaria, the vector is the anopheline mosquito and the disease-causing organism is the

malaria parasite.

Q. What is shell gland?

GULL

A gland-producing material, which forms the egg shell and is known as the shell gland, also opens into .this portion

Q. Example of hemiptera

Examples: Assassin bugs, Kissing bugs, Leaffooted bug, Lygus bug, Stink bug, Minute piratenbug,Big-eyed bug, Damsel bug

Q. Phthiraptera

Formally Anoplura and Mallophaga

Includes: Lice, Chewing Louse, Sucking Louse

Mouthparts: Chewing and Piercing-Sucking

Metamorphosis: Complete Metamorphosis

Order : Psocodea

Note: order or example examiner poch rha hy.

Q. Transmission

Transmission is the passing of a pathogen causing communicable disease from an infected host individual or group to a particular individual or group .Regardless of whether the other individual was previously infected

GULL

Q. Coxa

The coxa is the segment that articulates with the body, frequently partly lying in a more or less cupshaped hollow of the latter

It may be short or long and is generally freely movable on the body and powerful

Q. Cause of allergy

Allergy (hay fever, asthma, etc.) are frequently caused by exposure of susceptible persons to dried

insect scales and hairs

Q. Classes of Arthropods

1. Class #1- Hexapoda: Insects, six, jointed appendages

2. Class #2- Chilopoda- Centipedes, one pair of legs per body segment, multiple body segments

3. Class #3- Diplapoda- Millipedes, two pairs of legs per body segment, multiple body segments

4. Class #4- Crustacea- Crayfish, five pairs of jointed appendages

GULL

Q. Epidemiology

The study of disease in populations of an organism.

Q. Where live grasshopper of rice?

Various species of grasshopper are widely distributed in Nepal They are polyphagous and feed on leaves of rice, maize, millets, sugarcane, grasses etc

Q Fuzz purposes

Fuzz: It is used in production of mattresses, surgical cotton, photographic film and paper.

Q. Scientific name of early shoot borer

For Early shoot borer :

Trichogramma chilonis

Q.What is Biological Control?

Manipulation of biological organism to control pests by:

1- Release of predators/parasites/disease of an insect or weed Can be time consuming, expensive and difficult

Q.What is Host Plant Resistance?

Manipulating the crop to withstand or tolerate pests by

GULL

1- Natural breeding method

2- Genetically modified plants

Examples: Glandular-haired Alfalfa, Bt Corn,

Q.where Rice Grass hopper live 2

Various species of grasshopper are widely distributed in Nepal

They are polyphagous and feed on leaves of rice, maize, millets, sugarcane, grasses etc

Q.Which machinery is used to control pest 2

Uses machinery and/or other tools to control pests

1- Tillage: Tillage is the agricultural preparation of soil by mechanical agitation of various types, such as digging, stirring, and overturning

Q.Symptoms of damage of Fruit borer 3

Bolls showing regular, circular bore holes

Larvae seen feeding on the boll by thrusting their heads alone inside and leaving the rest of the body outside

Presence of granular faecal pellets outside the bore hole.

GULL

A single larva can damage 30-40 bolls.

Q.Why Agronomic practices that are designed to: 3

1-Optimize growing conditions for the crop

2-Anything that increases a crop's competitive edge will result in increased tolerance to pests often resulting in reduced pesticide use

Q- Create unfavorable conditions for the pest ?

1- Describe the places where heat treatment is used to kill the insects 3

2- Most insects cease reproduction and soon die at temperatures rom 100 to 1l0°F

3- The temperature of their surroundings should be raised to 120° or higher to kill them with heat artificially within a few hours

4- The length of period for effective treatment is more dependent upon the

a. time to heat the grain

b. clothing

c. other material in which the insects are located

GULL

5- Than upon the heat resistance of the pests themselves

6- Because no insect can survive more than an hour if actually exposed to a temperature of 120°F 7- Bales and bags are difficult to heat through

Q. Define Ovoviviparous.

Those having eggs that remain in the body of the parent for a time after hatching are termed ovoviviparous.

Q. What is meant by contact insecticides?

It was formerly supposed that certain materials called contact insecticides, which kill insects by contact with their bodies. It is caused death by entering the spiracles and closing them up, thus producing suffocation.

Q: What is seminal receptcal?

A side pouch called seminal receptacle is connected with the vagina is for the storage of the sperms which fertilize the eggs.

Q: Write down the scientific name of early shoot borer.
Trichogramma chilonis is the scientific name of early shoot borer.

GULL

Q: Define Arthopods.

The largest group of all, however, is the Arthropoda. Its members found in the seas, in fresh water, on land or even flying freely. A group with remarkable differences of structure and it is so abundant that all the other animals taken together are less than one-sixth as many as the arthropods.

Q: What is Biological Control?

Manipulation of biological organism to control pests by:

1- Release of predators/parasites/disease of an insect or weed

2- Can be time consuming, expensive and difficult

Q: Write down the Order of termites.

Blattodea is the Order of termites.

Q: What is the appearance of adult rice-ear?

The adult of Leptocorisa oratorius is green, light brown or mixed yellow in color.

Q: Write down the economically importance of insects.

GULL

1. Of the many thousands of insect species known to man some affect him and his property only slightly; some are beneficial; others are injurious in various degrees.

Q: Explain Positions Head of insects and its types.

The heads of different insects very much in form in the location of the mouth. In some cases this is on the underside; in others it is practically on the front.

Hypo-gnathous head perpendicular to longitudinal body axis for example Locust.

Pro- gnathous head extended to longitudinal body axis for example Weevils.

Opitho- gnathous head bent underneath thorax during rest for example Aphids.

Q. List the insect born diseases?

- Malaria
- Bubonic plague
- Typhoid
- Typhus Yellow fever

GULL

Q. Clean culture?

The destruction of vegetable trash and weeds is known as cultural clean

Q. Hopper burn cause?

Brown plant hopper is the most destructive pest of rice .. both nymph and adult feed on paddy, sugercan and grasses by sucking cell sap which turn the leaves yellow.

A heavy infestation produce symptoms of “hopper burn” i.e. leaves become dry and brown after insect feeding and patched of burned plants are often lodged.

Q.Damage cause by termits?

• They enter through cut ends of setts & feed on the soft tissue.

• The tunnel is filled with the soil.

• The termites attack setts, shoots, canes and also stubbles

• Entire shoot dries up and can be pulled out easily

Q. Physical control methods?

Physical factor controlling insects are following

GULL

- Heat
- Cold
- Light
- Moisture
- Electricity

Q. Enlist the plants require pollination?

Smyrna fig, clovers, melons, tomatoes, beans, peas,and most of common fruits require insects for their pollination

Q. Agronomic practice in cultural control?

1. Optimize growing conditions for the crop

2. Anything that increases a crop's competitive edge will result in increased tolerance to pests often resulting in reduced pesticide use

3. Create unfavorable conditions for the pest

Q. What is cotton?

Cotton is a soft, fluffy staple fiber that grows in boll, or protective Cotton (Gossypium spp), the king of fibers, usually commercial referred as white gold and one of the important commercial crops, plays a pivotal role in

human civilization, economic, political and social affairs of world.

Q. Economic threshold?

“Pest Population at which a grower must take action to prevent a pest populations fromreaching the economic injury level”

Q. Define pulvilus, pulvilli,empodium?

At the end of the last a pair of claws is generally found, and between them a sort of pad or cushion, the pulvillus

Sometimes there are three of these, in which case the outer ones are called the pulvilli and the middle one the empodium

Q. Intelligent cultivation?

Intelligent cultivation and use of fertilizers is well as pruning to remove injured or diseased parts will aid in insect control

Q. Arthropoda classes?

5 Classes of Arthropods

1. Class #1- Hexapoda: Insects, six, jointed appendages

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2. Class #2- Chilopoda Centipedes, one pair of legs per body segment, multiple body segments

3. Class #3- Diplapoda- Millipedes, two pairs of legs per body segment, multiple body segments

4. Class #4- Crustacea- Crayfish, five pairs of jointed appendages

5. Class #5- Arachnida- Spiders, four pairs of jointed appendages

Q. Insect pest management?

The stud of principles and experimental approaches to the science that underpins the development of working integrated pest management systems.

Q.Appearance of adult rice-ear.2

Rice ear- head bug

The adult of Leptocorisa oratorius is green, light brown or mixed yellow in color The adult are slender and about 20mm long.

Q. Define arthropods? 2

GULL

An “arthropod” is an invertebrate animal that has an exoskeleton, a segmented body, and jointed appendages any of a phylum (Arthropoda) of invertebrate animals (such as insects, arachnids, and crustaceans) that have a segmented body and jointed appendages, a usually chitinous exoskeleton molted at intervals, and a dorsal anterior brain connected to a ventral chain of ganglia

Q.Damage by termites?3

Termite

SYMPTOMS OF DAMAGE

They enter through cut ends of setts & feed on the soft tissue.

The tunnel is filled with the soil.

The termites attack setts, shoots, canes and also stubbles
Entire shoot dries up and can be pulled out easily

Q. What are attractant? 3

Attractants

1. Sometimes called attrahents are materials used to lure insects to traps and poisoned baits
2. The use of geraniol in Japanese beetle traps, sugar

sirups for ant baits, and fermenting sirup solutions fortrapping various moths are good examples.

Q. What is biological control?

Biological control or biocontrol is a method of controlling pests such as insects, mites, weeds and plant diseases using other organisms. It relies on predation, parasitism, herbivory, or other natural mechanisms, but typically also involves an active human management role.

ZOO506

LONG QUESTION

MIDTERM

QNO 1 Economic importance of insects?

ANSER

Antagonist Zoologist Group

Effort by Gull

A. Of the many thousands of insect species known to man some affect him and his property only slightly; some are beneficial; others are injurious in various degrees

B. Only rough estimates can be made of the monetary losses and gains by insect activities and even these are made with difficulty

C. Although it is the general practice to express losses in terms of dollar valuation

D. It probably would be safer to give them in bushels, tons, etc., because the relative values of different crops and products change from year to year

E. No crop of either vegetable or animal origin appears to be entirely free from insect injury

F. Field, truck and fruit of crops, both growing and in storage, household goods and food products, forests and the wood products derived from them, domestic animals and animal products, all are more or less liable to insect attack

G. Because there never has been a season free from the ravages of insects which might serve as a standard for comparison, much of this loss is not appreciated

H. If there could once be such a year entirely insect free, the difference would be apparent at once

QNO2 -How shot and top and internode borer can be biologically controlled?

ANSWER

Biological Control

- For Early shoot borer :

Trichogramma chilonis

- For Top borer :

Trichogramma chilonis or T. japonicum, Isotimajavensis

- For Internode borer :
  - Trichogramma chilonis
  - Cultivation of resistance variety
  - Irrigation at closer intervals for managing
  - Early shoot borer
  - Practice deep harvesting to destroy stubbles
  - Removal of water shoots to destroy
  - Top shoot borer

QNO3- Difference beween the exoskeleton of mollusks and arthropods?

ANSWER

In the possession of this external skeleton these animals have a seeming resemblance to the shells

(Mollusca) But the materials of which the skeleton is composed are quite different Being largely

calcium carbonate in the Mollusca And chitin which somewhat resembles horn in its nature,

sometimes with calcareous salts·deposited in it, in the Arthropoda

.The simple members of arthropod have bilaterally symmetrical body, though this condition is concealed somewhat by secondary changes in many of the group .The possession of a bilaterally symmetrical body consisting of a series of segments, an exoskeleton of chitin and the presence of jointed legs are, then, distinctive features of the arthropods .To separate the various groups of arthropods, other characters must be Used Aside from several small sections not often seen, there are five large and important divisions which call for recognitization

QNO4- Describe Metamorphosis.

ANSWER

In some of the simpler inflects, a few changes and growth only are needed to make them mature. These are therefore usually grouped together as the Ametabala, or insects having practically no metamorphosis.

The remaining insects, from this standpoint, form two groups: Hemimetabola, or Paurometabola: Those which

on hatching generally show some resemblance to the adults and reach maturity by a series of gradual changes.

Holometabola: And those which on hatching are totally unlike the adults and become adults by a somewhat different process.

Growth, however, is restricted because the insect is enclosed by a cuticula which, while elastic, to some extent, at least at its thiner places, has its limitations in this regard. Thus the insect is unable to reach its adult size within its cuticula, and a process called molting takes place. This is begun by a pouring out of fluid by the outside layer of living cells, the hypodermis, between it and the cuticula, separating the two Next a spilt appears somewhere in the cuticula usually along the back and the insect crawls out of its skin, i.e., molts.

QNO5- Pests of rice

ANSWER

PESTS OF RICE

Rice (Oryza sativa L.) is the most important staple food crop of the world

More than 60 % of worlds population depend on it for food.

Major insect pests

1-Brown Plant Hopper, Nilaparvata lugens (Homoptera: Delphacidae)

2-Yellow Rice Stem Borer, Scirpophaga incertulas (Lepidoptera: Pyralidae)

3-Rice Ear-head Bug, Leptocorisa oratorius (Hemiptera: Alydidae)

4-Rice Hispa, Dicladispa armigera (Coleoptera: Chrysomelidae)

5-Rice Grasshopper, Hieroglyphus banian (Orthoptera: Acrididae)

QNO6- Economically importance of insects

ANSWER

- The benefits to man from their activities are frequently overlooked in summing up the economic importance of insects

GULL

- Although in many cases it is impossible to say how much good is derived from insects 1-Attack on injurious species
- Reliable statistics are easily obtained relative to the production of useful insect products
- Those insects which attack injurious species probably form the most important group

They are divided into two sections:

1-The parasites that live in or on the different stages of other insects, then called hosts

2-The predators that capture and devour other insects

- Parasites usually, though not always, live in or on a single host insect during their entire development
- Whereas a single predator may derive its nourishment from many victim insects.
- Parasites are of many species, frequently so specialized in their habits that one kind of parasite attacks only one or a few host species
- In some cases the parasites work only in insect eggs and thus prevent any feeding loss by the host insect

- Other parasites attack early stages of the pests, in some instances killing them soon

QNO7-. Order coeloptera with example.

ANSWER

Coleoptera

Includes: Beetles and Weevils

1. Blister beetle
2. Boll weevil
3. Collops beetle
4. Darkling beetle
5. Dermestid beetle
6. Dung beetle
7. Fig beetle,
8. Flea beetle
9. Lady beetle
10. Long-horned beetle
11. Palo Verde root borer

GULL

12.Metallic wood borer

Mouthparts: Chewing, Weevil- Piercing Sucking

Metamorphosis: Complete Metamophosis

Beetles are a group of insects that form the order Coleoptera, in the superorder

Endopterygota. Their front pair of wings are hardened into wing- cases, elytra,

distinguishing them from most other insects. The Coleoptera, with about 400,000 species,

is the largest of all orders, constituting almost 40% of described insects and 25% of all known animal life-forms; new species are discovered frequently.

QNO8- Advantage of weeds destruction in cultural control?

ANSWER

1. 2- The destruction of vegetable trash and weeds, frequently called clean culture, is also an important factor, especially in the control of hibernating insects

2. Weeds not only interfere with successful crop growth by crowding, but they may reduce the vigor of the crop by competing for plant food in the soil

3. In winter they provide hiding places for hibernating pests

4. A pest appearing in the spring before its crop food plant is available may find certain weeds palatable until the crop has appeared

5. Other pests move to late summer and fall weeds after their particular crops have been harvested

QNO9- Explain Biological control?

ANSWER

1. Parasitic and predaceous insects as well as diseases of insects have already been discussed

2. As vital factors in the natural control of injurious insects

3. It is possible, however, for man to encourage such beneficial forms by introducing them into new areas

4. By growing them artificially for liberation whenever they become scarce in nature

5. Until recent years most of this work dealt with the introduction of parasites

6. From abroad to combat those foreign insect pests brought to this country accidentally but without their natural enemies

7. Certain native parasites and predators are now being reared in large numbers for liberation in areas badly infested with their host species

8. Probably the first introduction attempted from a foreign country was the now classic case of a ladybeetle brought from Australia to attack the cottony cushion scale

9. Which had already reached California and was endangering the existence of the citrus industry

10. Following this successful attempt many other parasites and predators have been imported

11. Which in many instances have proved valuable in reducing the numbers of foreign pests

12. Parasites should not be expected to kill off all their hosts

QNO10- Create unfavorable conditions for the peST?

ANSWER

1- Describe the places where heat treatment is used to kill the insects 3

2- Most insects cease reproduction and soon die at temperatures rom 100 to 1l0°F

3- The temperature of their surroundings should be raised to 120° or higher to kill them with heat artificially within a few hours

4- The length of period for effective treatment is more dependent upon the

a. time to heat the grain

b. clothing

c. other material in which the insects are located

5- Than upon the heat resistance of the pests themselves

6- Because no insect can survive more than an hour if actually exposed to a temperature of 120°F 7- Bales and bags are difficult to heat through.

THANK YOU

Antagonist Zoologist Group

Effort by Gull